

Sustainability Leadership

Activation Guide for Government, Business and Academia

A reference model and a practical process to deep
dive into sustainability competencies of the future

August 2025



Forward

Across countries, societies and sectors, one truth is clear: sustainable development is not progressing at the speed or scale our world demands. The cost is existential—yet the opportunity is extraordinary. Research indicates that steering economies onto a sustainable pathway could unlock around US\$12 trillion in new value every year²². Why, then, do we still lag? A critical gap is the absence of a shared, practical reference for sustainability leadership.

Over the past decades, the global agenda has focused heavily on measurement systems—sustainability reporting standards, responsible investment principles, climate and nature disclosures, and impact accounting frameworks. These have been essential for directing capital towards sustainable pathways. Yet, what is missing is equal attention to the collective capability of leadership—the ability of governments, businesses, and academia to act together, with competence, values, and vision, to accelerate the flow of resources and deliver transformative outcomes.

Though academia, consulting and multilateral organisations created frameworks and thought leaderships on sustainability leadership, a comprehensive movement on sustainability leadership and competency development is not visible as it is there for the measurement system and disclosures²⁶.

To close this gap, GCCOSS proposes a Sustainability Leadership Reference Model.

- The model defines leadership across two dimensions : Strategic–Operational and Analytic–Integrative
- Eight knowledge domains of Environmental; Social & Behavioural; Economics; Finance & Investment; Technology & Innovation; Governance, Policy & Legal; Decision Sciences; and Communication & Engagement.
- Equally important, the model captures eight defining characteristics of sustainable development: complexity and systemic thinking; transformational; intergenerational ; cross-border ; multi-stakeholder; ethics and values; adaptive resilience; and emotional intelligence.

The value of this model lies in providing a common reference—a language and framework that bridges sectors, disciplines, and cultures. It enables governments, business and academia to define what leadership for sustainability looks like in practice, align expectations across stakeholders, and build the collective capability to both accelerate progress and unlock the significant economic and societal opportunities ahead.

This is a call to move beyond measurement alone. By strengthening sustainability leadership as a lever, we can accelerate fund flows, close the implementation gap, and drive the systemic transformation, the world urgently needs.

Team GCCOSS



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Sustainable development is an existential challenge and incredible opportunity

Closing the leadership gap is urgent and critical, to drive systemic transformation

Current pledges inadequate to keep global temperature rise below 1.5°C.¹⁶

Atmospheric CO₂ has surpassed 420 ppm, breaching planetary boundaries.¹⁶

Climate and biodiversity loss as systemic threats that are on highest priority²³

69% decline in wildlife populations since 1970.²⁴



Intensifying existential challenge

A critical existential challenge is looming large at the confluence of environmental collapse and deepening social inequality. Planetary boundaries transgressed, biodiversity is plummeting, and we face perilous warming trajectories. Meanwhile, vast populations live on just \$2–7 daily, with rising income inequality, job uncertainty, and eroding social trust¹⁴. Climate impacts intensify these inequalities: poorer, marginalised groups suffer most—women farmers lose more during heatwaves and floods, and disadvantaged communities endure longer power outages amid disasters. Together, ecological breakdown and social inequity poses an existential challenge for the planet

Incredible Opportunity

Sustainable development presents significant economic and business opportunities. It drives innovation, opens new markets, and enhances competitiveness. By aligning with the Sustainable Development Goals (SDGs), businesses can access an estimated \$12 trillion in market opportunities across sectors like food, agriculture, cities, energy, and health²². Norges Bank's AI-driven investment approach shows how visionary leadership can strategically channel capital into climate mitigation and biodiversity protection, creating measurable planetary impact while generating positive fund performance.



Leadership action lagging

The leadership gap in sustainable development is a critical barrier to urgent action. Despite frameworks like the UN SDGs and the Paris Agreement, current leadership capacity is insufficient. The UN Global Compact stresses the shortage of leaders with systems thinking, ethical decision-making, and multi-stakeholder engagement skills¹⁹. This leadership gap undermines the speed and effectiveness of sustainability efforts.

Systemic Transformations

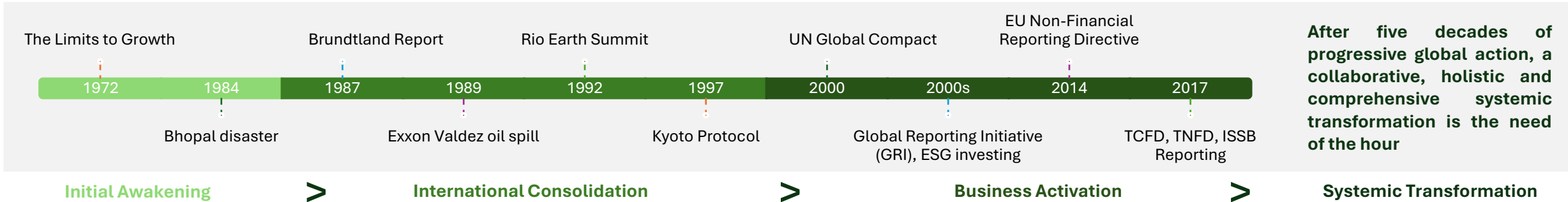
UN Report¹⁵ highlights the need for systemic transformations across energy, food, urban, and digital systems, which can create new business opportunities. The UNEP Report emphasises the importance of integrating financial, operational, communicative, and cultural dimensions to achieve sustainable development to foster economic resilience and long-term growth^{14,16}.

Multidimensional Response

Leaders today are also required to connect across domains—policy, finance, technology, and social systems—while keeping long-term impacts in focus. The UNDP Report highlights that the planetary crisis is not only environmental but also deeply social, demanding a multidimensional response that accounts for justice, inclusivity, and resilience¹⁴. Meanwhile, the UN Global Compact calls for cross-sector and multi-stakeholder collaboration as the only viable pathway to achieve transformational change at scale²⁰.

A common reference model for sustainability leadership is needed

To inspire and involve Government, Business and Academia, enabling Systemic Transformation



After five decades of progressive global action, a collaborative, holistic and comprehensive systemic transformation is the need of the hour

1 The Multidimensional Nature of Sustainability Leadership Requires Integrated Action

Sustainability challenges are inherently multidimensional; spanning climate, biodiversity, social equity, governance, and economic resilience. Yet leadership responses often remain fragmented. Traditional models tend to split into two camps: technical and financial leaders focusing narrowly on systems and metrics, and value-driven advocates emphasizing ethics and purpose. While each perspective is critical, operating in silos prevents the creation of transformational, integrated capabilities.

This “integration challenge” means that a centralized or single-lens approach is no longer sufficient. Leadership must instead be adaptive, multi-sectoral, and capable of translating knowledge into coordinated system-wide action. Without this shift, sustainability efforts will remain fragmented, incremental, and insufficient to address the scale of the crisis.

2 Environmental Activism to Sustainability Leadership

Over the last fifty years, sustainability leadership has undergone a profound transformation, evolving from grassroots environmental activism to a to an integrated multi-sector leadership imperative. Since then, sustainability leadership has expanded in scope and depth. It now integrates social equity, economic resilience, and resource stewardship (e.g., SDGs), fosters cross-sector collaboration (e.g., SBTi), and requires new professional competencies and technological enablement.

This historical progression underscores that sustainability leadership is no longer the domain of environmental activists alone. It has become a shared mandate, embedded in the structures and cultures of diverse organizations driven by both voluntary action and regulatory force. The future will require even deeper integration, measurable outcomes, and collaborative stewardship to accelerate progress at scale.

3 Need for a standard reference model

Sustainable transformation demands integrated responses from governments, investors, corporates, academia, regulators, and civil society. Governments are tasked with balancing growth, justice, and ecological integrity¹⁵ while corporates face various intensities of pressure from regulators, investors, standards setting bodies. Academics are emerging as a key player in creating capacities and competencies as well as nurturing cutting edge technologies .

The UN Global Compact stresses the shortage of leaders with systems thinking, ethical decision-making, and multi-stakeholder engagement skills.¹⁹ Similarly, the WEF Future of Jobs Report identifies leadership, social influence, and analytical skills as critical²³. However, there is no reference model that integrates these competencies across contexts and allows each player to deep dive on various topics and competencies to drive systemic transformation.

Sustainability Leadership a collective responsibility

Government, Business and Academia need to work together

Bridging Theory and Impact

A critical barrier to sustainability progress is the gap between scientific theory and real-world impact. This is where industry-academia bodies, can play a transformative role. These centers foster collaboration between scientific research and industry practice, ensuring cutting-edge knowledge informs practical solutions. By cultivating leadership competencies—analytical thinking, ethical decision-making, and social influence—they empower future leaders with systems thinking skills and multi-stakeholder engagement capabilities, vital for navigating complex challenges and driving meaningful change.



A collective endeavour

Building sustainability leaders is not the work of a single institution—it requires collective effort across academia, industry, and government, each contributing distinct but complementary roles.



The Power of Triple-Sector Partnerships :

No sector can act alone. Joint innovation hubs, policy think tanks, and leadership networks illustrate how triple sector collaboration pools resources and expertise. Such partnerships expose leaders to the full spectrum of sustainability challenges, ensuring leadership is widely available—not a privilege but a societal resource. In essence, building sustainability leaders is a shared responsibility—requiring intentional collaboration and continuous evolution to secure a future that is equitable, resilient, and thriving

Academia - Cultivating Systems Thinkers

Universities nurture future leaders by embedding cross-disciplinary systems thinking and ethics into education. Applied learning—through research partnerships, internships, and projects—ensures theory translates into real-world solutions.

Business -Turning Theory into Practice

Corporates provide the testing ground for sustainability leadership, where shifting markets and regulations demand innovation, resilience, and collaboration. Beyond technical skills, industry must invest in adaptive leadership capacities such as ethical leadership, change management and stakeholder engagement, while working closely with academia to align knowledge with practice.

Government – Creating Enabling Conditions

Governments shape the environment for leadership through policy, funding, and incentives. By convening academia and industry, the public sector fosters collaboration on systemic challenges like climate adaptation, waste management, water stewardship, nature based solutions and social equity.

A Reference Model can help activate the triple sector partnership

By widening the perspectives, align competencies and enable integrated action

2 dimensions and 4 quadrants

Two dimensions to be considered to understand at what level the knowledge domains to be used. The analysis will provide the specific skills and competence related to the knowledge domain based on the level and span of influence and control.



2

Knowledge Domains			
Environmental	Social & Behavioural	Economics	Finance & Investment
Technology & Innovation	Governance, Policy & Legal	Decision Sciences	Communication & Engagement

8 Knowledge Domains

The eight sustainability knowledge domains—environmental, social and behavioural, economics, finance and investment, technology and innovation, governance/policy/legal, decision sciences, and communication—offer a holistic foundation for leadership development. Structured across strategic, operational, and analytic integration, they enable leaders to address interconnected challenges. This framework builds essential skills and competencies to drive transformative, equitable, and resilient pathways to sustainable development.

3

Characteristics			
Complex & Systemic	Transformational	Intergenerational	Cross- Border
Multi-stakeholder	Ethical and Value Driven	Adaptive & Resilient	Emotional

8 Characteristics

Sustainability challenges are complex, systemic, intergenerational, and cross-border, requiring multi-stakeholder collaboration, ethical grounding, resilience, and emotional intelligence. By evaluating eight sustainability knowledge domains against these eight characteristics, this framework identifies essential attitudes, values, and soft skills. The approach equips leaders to navigate uncertainty, foster inclusivity, and drive transformational change, strengthening sustainability leadership development programs for a just and resilient future.

A Reference Model can help activate the triple sector partnership

By providing a multidimensional view of competencies that align organisational energy

The Idea Behind the Reference Model

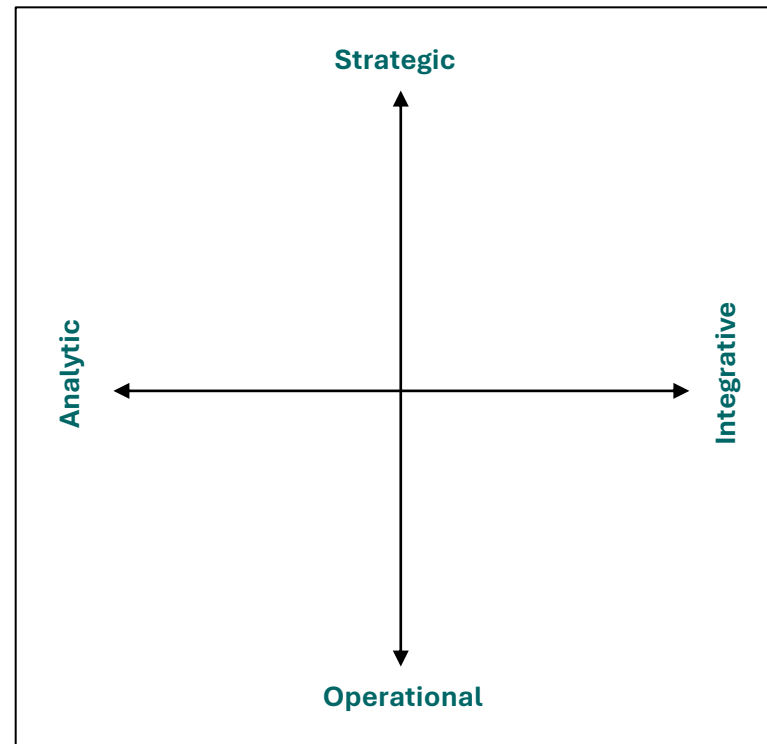
The reference model rests on a simple premise – sustainability leadership needs to address capabilities across two dimensions of strategic - operational and analytic - integrative. In an organisational context, these competencies need not rest with a single individual; these competencies must percolate across levels, functions and roles to create an organisational DNA of sustainable leadership.

First dimension: Strategic vs. Operational.

Strategic leadership is about the holistic, long-term view—envisioning sector-wide transformations, setting ambitious targets, and planning for decades through scenario modeling, capital allocation, and policy design.

Operational leadership ensures these visions are realized: executing initiatives, embedding standards in procurement, and tracking performance at business units, value chain as well product levels. Strategy mobilizes direction and resources, while operations deliver accountability and measurable outcomes.

The power of this axis lies in its balance: strategy sets the direction and mobilises resources for the long term, while operations ensure day-to-day discipline, innovation, and accountability. True sustainability leadership means excelling at both—connecting high-level ambition with on-the-ground action, and enabling governments, businesses, and academia to close the gap between theory and practice creating expected impact while preserving financial capital.



Second dimension: Analytical vs. Integrative.

Analytical leadership emphasizes evidence and systems thinking: using data science to assess current reality to identify interventions, and providing data-driven recommendations. Yet technical expertise alone is insufficient.

The integrative side unites diverse stakeholders—manufacturers, policymakers, recyclers, and consumers—into collective action. It involves navigating values, reconciling incentives, and crafting governance structures that enable collaboration. Numbers identify opportunities, but integrative leadership ensures shared ownership and long-term resilience.

Success depends on blending these skills: using analytics to illuminate opportunities and risks, while employing integrative leadership to foster cooperation and shared vision. Only by harnessing both dimensions can leaders drive an ecosystem transformation that is financially sustainable, environmentally resilient and socially adaptive.

A Reference Model can help activate the triple sector partnership

Through top down and value chain integration

Reference model explained

Practical relevance of reference model can be understood by considering two key transitions that are unfolding currently – net zero transformation of individual corporates and circular economy that is sustainable and resilient.

For example, developing and implementing a net zero strategy of a corporate entails translating a nation's net zero ambition into actionable results at the sector, company, and product levels. Strategically, leaders must start with the big picture: setting ambitious yet realistic emissions reduction targets in line with national commitments, envisioning sector-wide transformation, and identifying levers—policy changes, technology investments, value chain partnerships—that can drive the transition. This involves scenario planning, long-term capital allocation, and crafting a roadmap that not only navigates today's constraints, but also anticipates future opportunities and risks.

The operational dimension transforms strategy into measurable outcomes. For a company, this means executing decarbonisation initiatives: investing in renewable energy, retrofitting facilities, redesigning products, and embedding low-carbon procurement standards. It requires building systems to track the carbon footprint in real time—not only at the company level, but down to individual products and precursors in their supply chains. Operational excellence involves rigorous data management, cross-functional collaboration, and continual monitoring to ensure that every step—from raw material sourcing to product use by customers—contributes to the net zero goal.

The analytical versus integrative axis captures the dual mindset needed to tackle complex sustainability challenges—like building a circular economy ecosystem. Analytical thinking is essential for mapping material flows: tracking where resources come from, how they are used, and where they end up after use. It involves collecting and interpreting data on waste streams, identifying points where materials can be recycled, reused, refurbished, reformulated, or remanufactured, and quantifying the efficiency and impact of these interventions. Analytical leaders excel at breaking down problems, applying systems thinking, and providing evidence-based recommendations.

Yet, numbers and data alone cannot drive true circularity. The integrative dimension is equally vital. Creating a thriving circular economy is not just about efficiency; it is about orchestrating collaboration among diverse ecosystem players—manufacturers, recyclers, regulators, consumers, innovators—whose interests may not always align. Integrative leaders bring these stakeholders together, navigating differences in values, incentives, and risk appetites. They identify the gaps and opportunities within the regulatory landscape, negotiate supportive policies, and design risk-and-reward models that encourage collective action and strike win – win partnerships..



A Reference Model can help activate the triple sector partnership

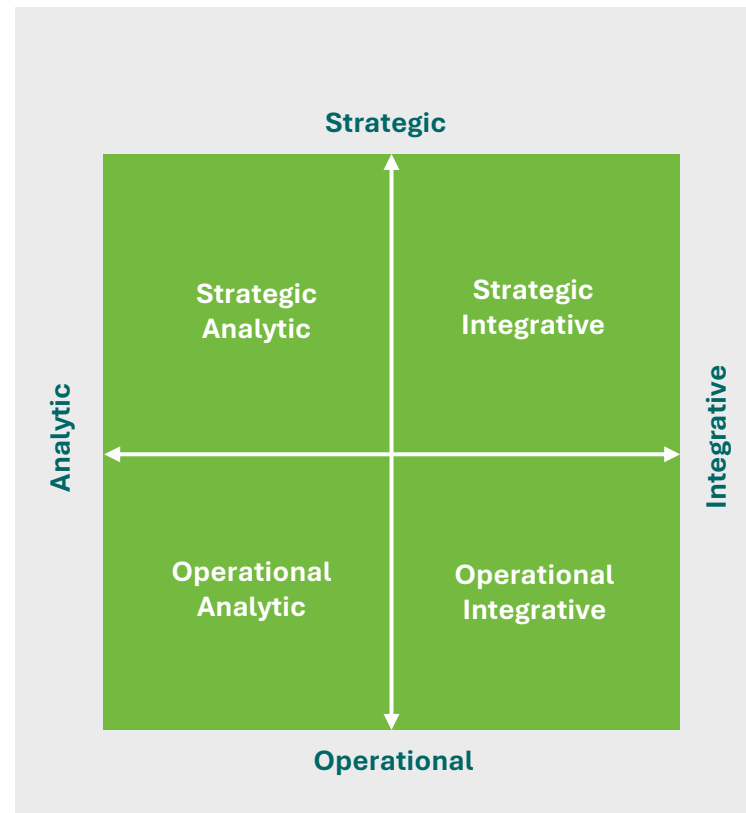
Place leadership priorities in an integrated manner to orchestrate thought and action

Reference model explained

The reference model also provide an opportunity to understand four leadership archetypes that can be mapped to organisational roles. The four quadrants are explained below.

Strategic Analysts is an archetype that suits CXOs, who understand the systemic interconnections between environmental and social that impacts economic, fiscal and monetary systems. Leaders in this space understands how natural systems interacts with human systems. They can read and interpret economic models and policy arguments that impacts business models and investment decisions. They think in terms of trade-offs and long term system impacts of the emerging environmental and social risks and incorporate the mitigation measures into strategy development and implementation

Analytical Operations This is where is sustainability performance is measured and performance aspirations are translated into investment decisions and monitoring, leveraging technology. This archetype is an expert in triple bottomline and has an in depth understanding of sustainable finance tools like green bonds, carbon markets, plastic credits, nature markets, climate funds, and blended finance models. They keep up with innovations — from renewable energy and AI for climate modelling to new materials and circular economy solutions. This is the engine room of sustainable development — it's where ideas get built, funded, and scaled.



Strategic Integrative is the ability to look far into the future and across boundaries to engage and influence the ecosystem strategically. Leadership archetype in this quadrant will ask: “How will this decision affect people 50 years from now? And will value cultural continuity and diversity, knowing that different generations and communities bring different perspectives. They communicate long-term benefits in a way that feels real and relevant today. Such Intergenerational leaders keeps us anchored to the fact that sustainability is about passing on a thriving planet to those who will come after us.

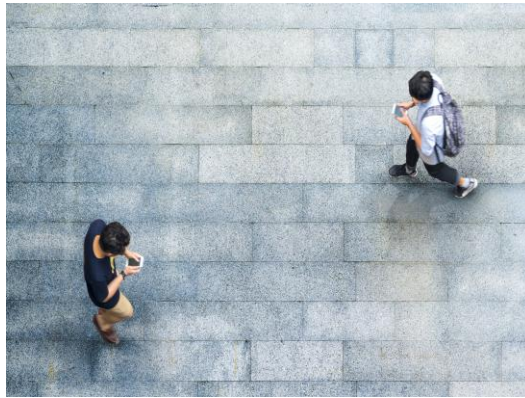
Operational Integrative is the people side of sustainability — leading through uncertainty, conflict, and change and ensures operational integration across boundaries happen. They inspire others to act, even when the path is unclear and work with stakeholders who have very different interests and values, finding common ground. Are resilient — able to adapt when conditions change, whether due to a natural disaster, a political shift, or new scientific findings. Keep ethics at the core, ensuring that sustainability actions are fair and inclusive..This quadrant is about building trust, aligning interests, and mobilising action across diverse groups.

Reference Model helps Government, Business and Academia to focus and deep dive

By identifying topic specific competencies to address specific characteristics of sustainable development

Sustainability leadership must rest on a strong understanding of environmental systems, social dynamics, and economic realities. Environmental sciences help leaders grasp planetary boundaries and resilience needs, while social and behavioural sciences illuminate human choices, values, and collective action. Economics provides frameworks for equitable growth, while finance and investment mobilise capital toward impactful solutions. Together, these disciplines ensure leaders can balance ecological integrity with human well-being and economic prosperity. A sustainability leader must translate scientific evidence and financial mechanisms into actionable strategies, driving transformative change while securing long-term value for stakeholders and society.

Environmental sciences equip leaders with knowledge of climate systems, ecosystems, and resource management. This domain cultivates skills to assess environmental risks, design mitigation strategies, and integrate ecological considerations into decision-making. It grounds leadership in planetary boundaries, ensuring strategies respect ecological limits while enabling regenerative and resilient development.



Economics provides tools to understand resource allocation, incentives, and trade-offs. For sustainability leadership, it emphasises integrating externalities, valuing natural and social capital, and promoting circular models. Leaders gain competencies to balance growth with equity and environmental limits, ensuring that long-term value creation aligns with both planetary health and societal well-being.



Social and Behavioural Sciences highlights human behaviour, cultural dynamics, and social equity. Leaders shall understand stakeholder motivations, community engagement, and social justice. By integrating behavioural insights into strategies, leaders can influence sustainable choices, foster inclusion, and build trust across diverse groups, ensuring solutions are socially acceptable, equitable, and ethically grounded.

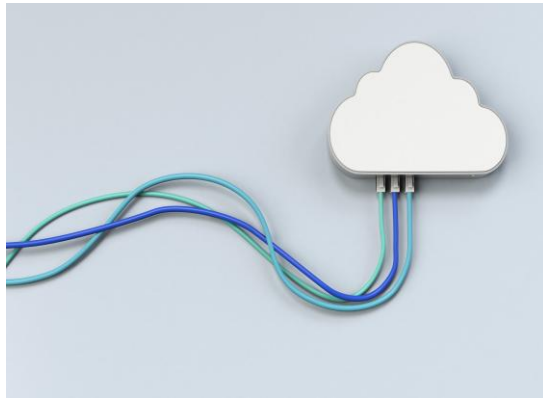


Finance and Investment domain links capital flows with sustainability priorities. Leaders must understand green finance, ESG integration, impact investing, and risk disclosure. Finance and investment knowledge enables leaders to mobilise resources for sustainable projects, align corporate funding with global goals, and leverage markets to accelerate transitions towards low-carbon and socially responsible futures.

Reference Model helps Government, Business and Academia to focus and deep dive

By identifying topic specific competencies to address specific characteristics of sustainable development

Technology and innovation are critical enablers of sustainable transformation, unlocking pathways from clean energy to circular economies. Governance, policy, and legal frameworks provide the guardrails for accountability, fairness, and systemic change. Decision sciences bring analytical tools for managing complexity and uncertainty, while communication bridges knowledge and action across diverse stakeholders. For sustainability leadership, these domains form the backbone of informed, transparent, and future-oriented decisions. Leaders must integrate innovation with governance, science with law, and evidence with dialogue—ensuring that progress is accelerated responsibly. Together, they create the foundation for scaling solutions across industries and societies.



Technology and innovation drive scalable sustainability solutions. Leaders must understand digital, clean, and disruptive technologies that enable efficiency, transparency, and resilience. This domain fosters competencies in evaluating technological trade-offs, encouraging innovation ecosystems, and aligning technological progress with ethical and environmental standards, positioning leaders to drive transformative and future-ready strategies.

Governance, policy, and legal frameworks shape sustainability transitions. Leaders must navigate regulations, corporate governance, and global agreements. This domain develops competencies in compliance, advocacy, and ethical leadership, ensuring accountability and transparency. It equips leaders to influence policy, align organisational practices with evolving norms, and build legitimacy in sustainability agendas.



Communication and Leadership is the corner stone of effective leadership. This domain emphasises narrative-building, stakeholder dialogue, and change management. Leaders must inspire action, translate complex issues, and foster trust. By mastering communication, sustainability leaders can mobilise teams, influence stakeholders, and embed sustainability principles into organisational culture and public discourse for lasting impact.



Decision sciences provide tools for evidence-based, data-driven leadership. Leaders must apply systems thinking, modelling, risk analysis, and scenario planning to complex sustainability challenges. This domain develops analytical rigour, enabling leaders to balance uncertainty with foresight, evaluate trade-offs, and make transparent, informed choices that advance long-term resilience and organisational sustainability outcomes.

Reference Model helps Government, Business and Academia to focus and deep dive

By identifying topic specific competencies to address specific characteristics of sustainable development

Sustainable development challenges are inherently complex, systemic, and interconnected across geographies and generations. Leadership requires the ability to embrace this complexity, think systemically, and design interventions that transform entrenched patterns. Intergenerational perspectives remind us that today's actions—or inactions—will shape opportunities for future generations. Cross-border dynamics highlight the global interdependence of resources, supply chains, and climate impacts. Leaders who can navigate these four characteristics transcend short termism and nationalism to champion policies, strategies, and collaborations that reflect the realities of a shared planet. These capabilities anchor sustainability leadership in foresight, bold action, and global cooperation.

Sustainability challenges are **complex and systemic** across ecological, social, and economic systems. Leaders must adopt systems thinking to recognise interdependencies, avoid unintended consequences, and design holistic solutions. This characteristic develops the ability to navigate complexity, anticipate ripple effects, and foster integrated strategies essential for resilient and sustainable transformation.



Sustainable development is **intergenerational**. Leaders must integrate long-term perspectives into decision-making, balancing short-term pressures with enduring outcomes. This characteristic nurtures foresight, stewardship, and accountability, enabling leaders to safeguard planetary resources, uphold equity across generations, and embed sustainability principles into organisational strategy and policy frameworks.



Sustainability requires **transformations**, not incremental adjustments. Leaders must drive transformative change in culture, strategy, and operations. This characteristic cultivates vision, innovation, and courage to challenge conventional models. It equips leaders to champion disruptive solutions, align organisations with sustainability imperatives, and inspire collective action toward large-scale systemic transformation.



Environmental and social issues are **cross-border**. Leaders must collaborate across geographies, cultures, and regulatory contexts. This characteristic builds global awareness, cross-cultural sensitivity, and the ability to manage international interdependencies. It prepares leaders to engage in transnational partnerships, address global risks, and harmonise local strategies with global sustainability agendas.

Reference Model helps Government, Business and Academia to focus and deep dive

By identifying topic specific competencies to address specific characteristics of sustainable development

Effective sustainability leadership is not only technical but deeply human. It requires multi-stakeholder collaboration, where diverse actors align around common goals. Ethics and values provide the moral compass to ensure that choices uphold justice, equity, and planetary stewardship. Adaptability and resilience prepare leaders to thrive amid uncertainty, disruption, and crises. Emotional intelligence enables empathy, trust-building, and inclusive engagement. These soft competencies complement technical expertise, shaping leaders who can inspire, mobilise, and sustain collective action. Embedding these traits ensures that sustainable development leadership is values-driven, people centred, and capable of delivering long-term, systemic change.



Sustainability leadership rests on **ethical and value driven** decision making. Leaders must ground actions in integrity, fairness, and respect for life. This characteristic reinforces ethical reasoning, transparency, and purpose-driven leadership. It ensures sustainability decisions align with universal values, enhance legitimacy, and build trust among stakeholders in complex organisational contexts.



Sustainability needs **emotional intelligence** to have empathy, self-awareness, and relational skills. Leaders must understand human emotions to inspire, engage, and unite diverse stakeholders. This characteristic cultivates compassion, active listening, and constructive conflict management. Emotional intelligence enables leaders to build trust, mobilise collective will, and foster inclusive cultures essential for sustainable transformation.

Sustainability requires the **multi-stakeholder** engagement of governments, businesses, communities, and civil society. Leaders must build inclusive processes that balance diverse interests and perspectives. This characteristic strengthens dialogue, negotiation, and coalition-building skills, equipping leaders to manage conflicts, foster shared ownership, and co-create solutions that reflect collective aspirations and responsibilities.



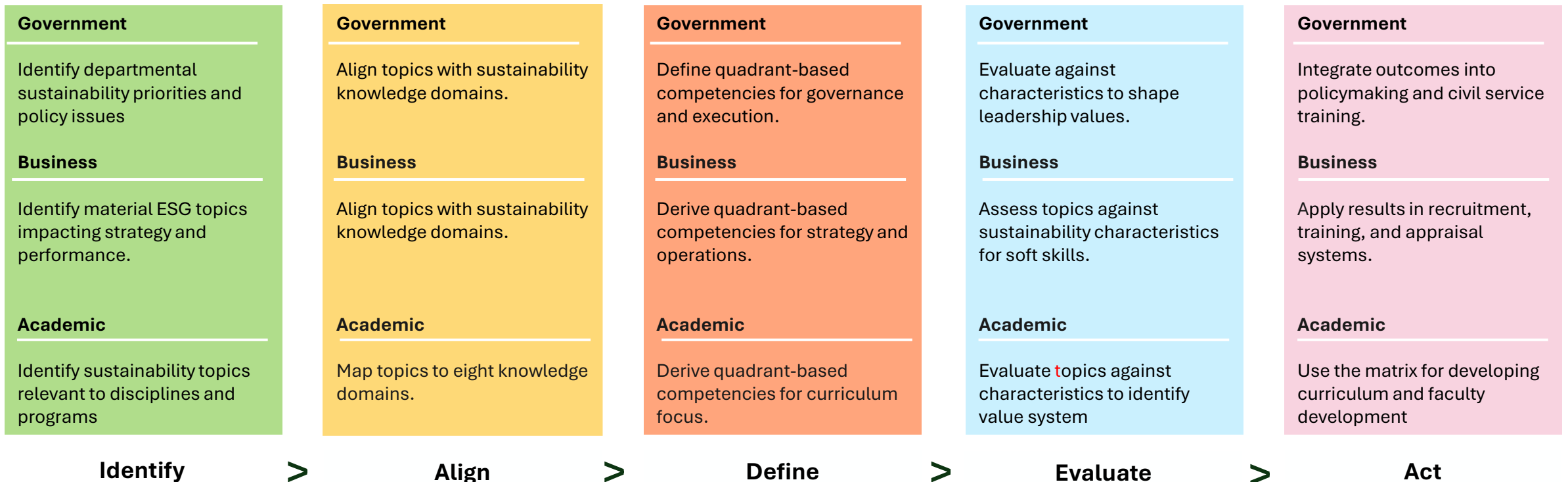
Sustainability leaders need **adaptability and resilience** to face volatility, uncertainty, and crises. They must adapt strategies while maintaining long-term focus. This characteristic develops resilience, agility, and problem-solving capacity. It equips leaders to manage shocks, pivot effectively, and cultivate organisational cultures capable of thriving under disruption, ensuring sustained progress in dynamic environments.



A systematic materiality based process unravel specific competencies

By mapping knowledge domains against the quadrants and characteristics

The proposed reference model provides a systematic approach to developing sustainability leadership competencies by integrating technical knowledge and values-based attributes. Organisations begin by identifying material topics most relevant to their context—businesses may use sustainability reporting materiality matrices, governments focus on departmental priorities, and academia emphasises curriculum needs. These topics are mapped against eight sustainability knowledge domains to ensure comprehensive coverage of environmental, social, economic, financial, technological, governance, decision science, and communication dimensions. Next, topics are analysed within four quadrants—Strategic-Analytical, Strategic-Integrative, Operational-Analytical, and Operational-Integrative—to derive technical competence statements. Parallely, topics are assessed against eight sustainability characteristics—complexity, transformational, intergenerational, cross-border, multi-stakeholder, ethics, adaptability, and emotional intelligence—to identify values, attitudes, and soft skills. The resulting competence matrices can then be applied across organisational systems: for businesses, in recruitment, training, and appraisal; for governments, in policy capacity building; and for academia, in curriculum design and professional education.



Where to focus and deepen

Urgent priority on financial and technological proficiency

Where to Focus Now

Sustainability today is both a sprint and a marathon. Climate shocks, resource pressures, shifting regulations, and stakeholder demands make the call for effective leadership urgent. But where should leaders focus to drive meaningful change?

Two skillsets stand out:

- **Financial & Technological Proficiency** – Leaders must link ambition with financial realism and technological savvy, from carbon markets to digital platforms. Those who bridge vision with practical levers of capital and innovation will drive the biggest impact.
- **Transformational Leadership** – Beyond strategy, leaders must inspire trust, mobilise coalitions, and dismantle silos. They empower others, adapt under pressure, and rally diverse actors around shared purpose.

Emerging Priority: Intergenerational Vision : True sustainability looks beyond short-term gains. Leaders must ask: will today's solutions endure? Are we preparing future generations to thrive? This requires humility and openness to younger voices.

Foundation: Systems & Economics Mastery : Sustainability leadership must rest on understanding how environmental, social, and economic systems interact. Bridging this expertise into practice helps identify leverage points and anticipate unintended consequences.

Path Forward :The focus now: sharpen financial and technological agility, elevate transformational leadership, embed intergenerational thinking, and translate systems knowledge into action. Only then can leaders meet today's urgency while laying foundations for lasting prosperity.



Where to focus and deepen

By mapping knowledge domains against the quadrants and characteristics

**Deepening the Skills:**

In today's fast-changing sustainability landscape, leadership requires skills that are robust, adaptive, and deeply interconnected. The dual-matrix framework strengthens these skills by mapping seven core knowledge domains across strategic–operational and analytical–integrative dimensions, offering a systematic yet flexible approach.

Addressing Complexity and Integration:

Skills must extend beyond technical expertise to include systems thinking, cross-sector solutions, and ethical decision-making that reflect sustainability's interdependencies. Leadership effectiveness will largely be based on the value system.

Strategic and Operational Alignment:

Leaders at different levels need complementary skills—from scenario analysis and systems thinking at the strategic level to metrics, performance indicators, and collaboration at the operational level.

Utility for Government, Business, and Regulators:

The framework guides role clarity, recruitment, re-skilling, and cross-functional collaboration, helping organizations identify gaps and strengthen adaptive capacity.

Transformative Impact for Academia:

It supports curriculum design and interdisciplinary research, equipping future leaders with systemic thinking, adaptability, value-based judgment, and intergenerational responsibility. It can also inform faculty development programs.

Enhancing Stakeholder Engagement:

The framework creates a common language across sectors, enabling benchmarking, informed policymaking, and collective action.

Adaptability and Customization:

Its flexibility allows organizations and academic institutions to refine it to their own context, ensuring relevance and future-proof leadership development.

By deepening skills across these dimensions, leaders move beyond fragmented expertise toward integrated, transformative capabilities—essential for shaping the future of sustainability.

A systematic materiality based process unravel specific competencies

By mapping knowledge domains against the quadrants and characteristics – Sample output of analysis

Knowledge Domain	Strategic – Analytical	Strategic – Integrative	Operational – Analytical	Operational – Integrative
1. Environmental Sciences	Assess climate risks and environmental trends for long-term strategy	Embed ecological principles into corporate or national strategy	Conduct impact assessments, carbon footprinting	Implement sustainability practices in operations (waste, water, energy)
2. Social & Behavioural Sciences	Analyze demographic and behavioral trends shaping sustainability	Integrate social inclusion, equity, and diversity into strategy	Assess stakeholder needs, surveys, community profiling	Execute programs for diversity, inclusion, employee well-being
3. Economics	Model sustainable growth pathways, cost-benefit tradeoffs	Embed sustainable development goals into macro/micro-economic policies	Perform economic impact studies	Implement sustainable procurement and circular economy practices
4. Finance & Investment	Evaluate ESG risks and opportunities for investment strategy	Integrate sustainability into capital allocation and financing models	Assess project-level financial sustainability	Execute sustainable finance mechanisms (green bonds, ESG reporting)
5. Technology & Innovation	Forecast disruptive sustainable technologies for strategic advantage	Embed innovation in sustainability transitions and partnerships	Conduct technology feasibility and lifecycle analyses	Implement digital/tech-enabled sustainability solutions
6. Governance, Policy & Legal	Analyze global and national policy trends, compliance risks	Integrate governance frameworks into sustainability strategy	Evaluate regulatory compliance and reporting requirements	Implement governance processes, legal compliance, accountability systems
7. Decision Sciences	Use systems modeling and scenario planning for sustainability strategies	Integrate data-driven decision support into leadership practices	Conduct operational analytics and optimization	Apply decision tools in project execution and daily operations
8. Communication & Engagement	Analyze stakeholder sentiment and influence on long-term strategy	Integrate multi-stakeholder collaboration into sustainability plans	Assess communication effectiveness with data and metrics	Implement engagement campaigns, change management, and dialogue platforms

A systematic materiality based process unravel specific competencies

By mapping knowledge domains against the quadrants and characteristics – sample output of analysis

Knowledge Domain	Complex & Systemic	Transformational	Intergenerational	Cross-Border	Multi-stakeholder	Ethics & Value-driven	Adaptive & Resilient	Emotionally Intelligent
1. Environmental Sciences	Systems thinking on ecosystems	Driving ecological shifts	Stewardship for future generations	Global ecological awareness	Collaborating across science & communities	Commitment to environmental ethics	Resilience to ecological shocks	Empathy for nature & people
2. Social & Behavioural Sciences	Understanding social complexity	Driving social change	Promoting equity for future generations	Cultural sensitivity	Engaging diverse communities	Justice and fairness	Adapting to social shifts	Empathy, inclusivity
3. Economics	Holistic economic systems view	Driving sustainable economic reform	Long-term economic planning	Global economic cooperation	Multi-sectoral economic dialogue	Fairness in resource allocation	Resilient economic design	Balanced stakeholder sensitivity
4. Finance & Investment	Systemic view of capital flows	Transforming finance for sustainability	Long-term investment horizon	Global finance networks	Multi-stakeholder ESG investment	Ethical investment principles	Resilience in financial shocks	Trust-building with investors
5. Technology & Innovation	Systems innovation mindset	Disruptive sustainable technologies	Future-focused innovation	Global technology sharing	Collaborative innovation	Tech ethics	Adaptability to new technologies	Human-centered innovation
6. Governance, Policy & Legal	Integrated governance systems	Policy innovation	Laws for intergenerational equity	Cross-border governance collaboration	Engaging civil society & businesses	Justice and accountability	Adaptive governance	Integrity, fairness
7. Decision Sciences	Systemic problem solving	Transformational decision-making	Long-term scenario planning	Cross-border decision frameworks	Collaborative decision models	Evidence-based ethics	Resilient decision systems	Active listening, fairness
8. Communication & Engagement	Systemic communication flows	Transforming narratives	Educating future generations	Cross-cultural communication	Multi-stakeholder dialogue	Honest, transparent communication	Adaptive messaging	Empathy-driven communication

Building collective leadership across Government, Business and Academia

An urgent need to future prosperity

Why This Matters Now More Than Ever

Change is accelerating—climate impacts, technology shifts, social movements, and geopolitics collide unpredictably. Sustainability leaders must be:

- Quick learners who adapt fast.
- Bridge-builders across people, sectors, and ideas.
- Long-term thinkers who safeguard tomorrow.

The Sustainability Leadership Framework isn't the only model, but it offers a practical lens to see the full spectrum of skills and the balance leaders must strike.

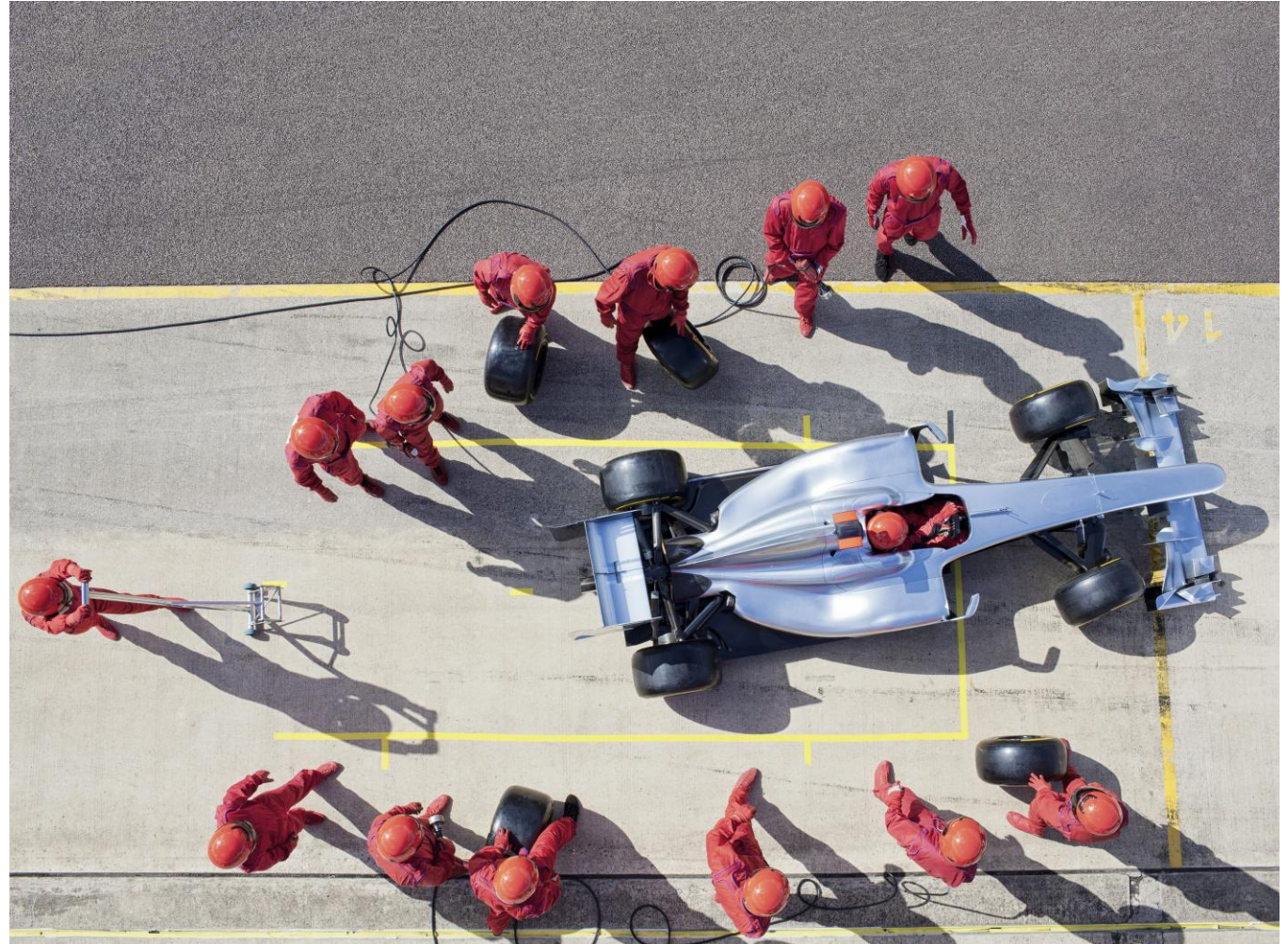
From Individuals to Teams

No single leader excels in all quadrants. Some think in systems but struggle to inspire; others embrace technology but find cultural change harder. The real power lies in balanced teams:

- One systems thinker,
- One visionary storyteller,
- One financial/tech expert,
- One stakeholder engager.

Together, they unlock transformation.

Sustainable development is a “wicked problem”—complex but not impossible. The Framework helps identify the right leadership for each challenge, build teams with complementary strengths, and develop missing skills. In short, it turns good intentions into coordinated, effective action—exactly what the sustainability movement needs today.



Building collective leadership across Government, Business and Academia

An urgent need to future prosperity

The urgency of sustainable development cannot be overstated. Despite progress in disclosure standards and financial reporting, the world is still far from the trajectory required to avoid an existential crisis. The GCCOSS Sustainability Leadership Reference Model provides a practical framework for building the competencies and values needed to accelerate this transition. Now, the task before us is action.

For Individuals:

Embrace sustainability leadership as a personal responsibility. Use the reference model to self-assess strengths and gaps in knowledge and soft skills. Invest in continuous learning—whether in environmental sciences, financial literacy, or emotional intelligence—and adopt values of resilience, ethics, and collaboration in your daily choices.

For Governments:

Apply the model to design leadership programs for public officials, ensuring decision-making reflects systemic, cross-border, and intergenerational perspectives. Embed competence matrices into departmental performance systems. Prioritise partnerships with academia and business to co-create sustainable policies that are both evidence-driven and inclusive.

For Businesses:

Go beyond reporting by embedding the competence framework into recruitment, training, and appraisal systems. Use the reference model to align sustainability materiality with leadership skills across strategy, operations, and culture. Develop cross-sector collaborations that unlock innovation, accelerate decarbonisation, and create new market value.

For Academia:

Incorporate the reference model into curricula, research agendas, and capacity-building initiatives. Equip future leaders with the interdisciplinary knowledge and values needed to lead systemic change. Facilitate dialogue platforms where students, businesses, and governments co-develop solutions to sustainability challenges.

This is a collective journey. By embracing this reference model, individuals, institutions, and sectors can deepen skills, foster partnerships, and upgrade ecosystem-wide competencies. Together, we can shift from fragmented efforts to a unified leadership movement capable of steering the world toward sustainable prosperity.



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